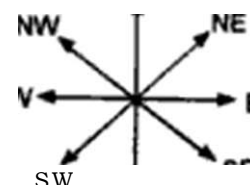


8. DIRECTION SENSE TEST

In this test, the questions consist of a sort of direction puzzle. A successive follow-up of directions is formulated and the candidate is required to ascertain the final direction or the distance between two points. The test is meant to judge the candidate's ability to trace and follow correctly and sense the direction correctly.

The adjoining figure shows the four main directions (North N, South S, East E, West W) and four cardinal directions (North East NE, North West NW, South East SE, South West SW) to help the candidates know the directions.



ILLUSTRATIVE EXAMPLES

Ex* 1. A man is facing west. He turns 45° in the clockwise direction and then another 180° in the same direction and then 270° in the anticlockwise direction. Which direction is he facing now ? **(Hotel Management, 1997)**

- (a) South (b) North-west (c) West (d) South-west

Sol. Clearly, the man initially faces in the direction OA. On moving 45° clockwise, he faces in the direction OB. On further moving 180° clockwise, he faces in the direction OC. Finally, on moving 270° anticlockwise, he faces in the direction OD, which is South-west. Hence, the answer is (d).

B*

D*'

C*

Ex. 2. One day, Ravi left home and cycled 10 km southwards, turned right and cycled 5 km and turned right and cycled 10 km and turned left and cycled 10 km. How many kilometres will he have to cycle to reach his home straight ? **(Assistant Grade, 1995)**

- (a) 10 km (b) 15 km (c) 20 km (d) 25 km

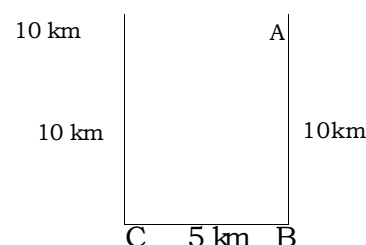
Sol. Clearly, Ravi starts from home at A, moves 10 km southwards upto B, turns right and moves 5 km upto C, turns right again and moves 10 km upto D and finally turns left and moves 10 km upto E.

Thus, his distance from initial position A

$$= AE = AD + DE$$

$$= BC + DE = (5 + 10) \text{ km} = 15 \text{ km}.$$

Hence, the answer is (b).



Ex. 3. A child is looking for his father. He went 90 metres in the east before turning to his right. He went 20 metres before turning to his right again to look for his father at his uncle's place 30 metres from this point. His father was not

there. From there, he went 100 metres to his north before meeting his father in a street. How far did the son meet his father from starting point ?

(Central Excise, 1906)

- (a) 80 metres (b) 100 metres (c) 140 metres (d) 260 metres

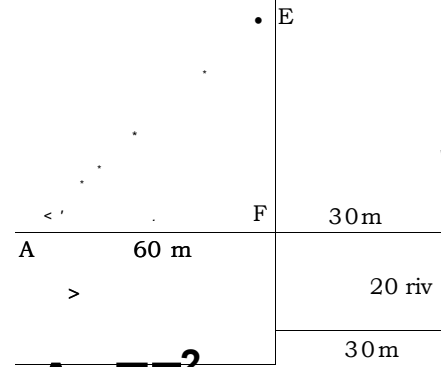
Sol. Clearly, the child moves from A 90 m eastwards upto B, then turns right and moves 20 m upto C, then turns right and moves 30 m upto D. Finally, he turns right and moves 100 m upto E.

Clearly, $AB = 90$ m, $BF = CD = 30$ m.

So, $AF = AB - BF = 60$ m.

Also, $DE = 100$ m, $DF = BC = 20$ m.

So, $EF = DE - DF = 80$ m.



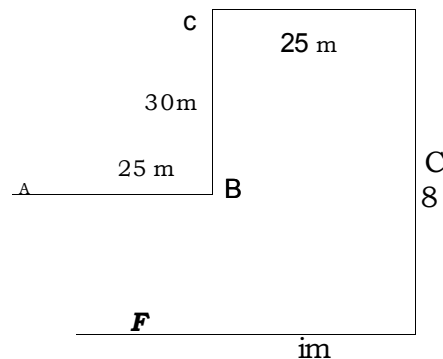
$$\begin{aligned} \text{His distance from starting point } A &= AE = \sqrt{AF^2 + EF^2} \\ &= \sqrt{(60)^2 + (80)^2} \\ &= \sqrt{3600 + 6400} = \sqrt{10000} = 100 \text{ m.} \end{aligned}$$

Hence, the answer is (b).

Ex. 4. Kailash faces towards north. Turning to his right, he walks 25 metres. He then turns to his left and walks 30 metres. Next, he moves 25 metres to his right. He then turns to his right again and walks 55 metres. Finally, he turns to the right and moves 40 metres. In which direction is he now from his starting point ?

- (a) South-west (b) South (c) North-west (d) South-east

Sol. Kailash turns towards right from north direction. So, he walks 25 m towards east upto B, turns left and moves 30 m upto C, turns right and goes 25 m upto D. At D, he turns to right towards the south and walks 55 m upto E. Next, he again turns to right and walks 40 m upto F, which is his final position. F is to the south-east of A. So, he is to the south-east from his starting point. Hence, the answer is (d).



Ex. 5. Deepa moved a distance of 75 metres towards the north. She then turned to the left and walking for about 25 metres, turned left again and walked 80 metres. Finally, she turned to the right at an angle of 45° . In which direction was she moving finally ?

- (a) North-east (b) North-west (c) South
(d) South-east (e) South-west

Sol. Deepa started from A, moved 75 m upto B, turned left and walked 25 m upto C. She then turned left again and moved 80 m upto D. Turning to the right at an angle of 45° she was finally moving in the direction DE i.e., South-west.

Hence, the answer is (e).

80 m 75 m



Ex. 6. Kunal walks 10 km towards North. From there he walks 6 km towards South. Then, he walks 3 km towards East. How far and in which direction is he with reference to his starting point ? **(M.B.A 1996)**

- (a) 6 km West . (b) 7 km West (c) 7 km East (d) 5 km North-East

Sol. Clearly, Kunal moves from A 10 km northwards upto B, then moves 6 km southwards upto C, turns towards East and walks 3 km upto D.

Then, $AC = (AB - BC) = (10 - 6) = 4 \text{ km};$

$CD = 3 \text{ km}.$

So, Kunal's distance from starting point A

$$= AD = \sqrt{AC^2 + CD^2} = \sqrt{4^2 + 3^2} = 5 \text{ km}.$$

Also, D is to the North-east of A.

Hence, the answer is (d).

Ex. 7. Johnson left for his office in his car. He drove 16 km towards north and then 10 km towards west. He then turned to the south and covered 5 km. Further, he turned to the east and moved 8 km. Finally, he turned right and drove 10 km. How far and in which direction is he from his starting point ?

- (a) 2 km West (b) 5 km East (c) 3 km North
(d) 6 km South (e) None of these

Sol. Clearly, Johnson drove 16 km from A to B northwards and then 10 km from B to C towards west. He then moves 5 km southwards from C to D and 8 km eastwards upto E. Finally, he turned right and moved 10 km upto F.

\ A and F lie in the same straight line and F lies to the west of A.

So, Johnson's distance from the starting point A = AF = (BC - DE) = (10 - 8) km = 2 km.

Hence, the answer is (a).

EXERCISE 8A

- A man is facing south. He turns 135° in the anticlockwise direction and then 180° in the clockwise direction. Which direction is he facing now ?
(a) North-east (b) North-west (c) South-east (d) South-west
- A man is facing north-west. He turns 90° in the clockwise direction and then 135° in the anticlockwise direction. Which direction is he facing now ?
(a) East (b) West (c) North (d) South
- A man is facing north-west. He turns 90° in the clockwise direction, then 180° in the anticlockwise direction and then another 90° in the same direction. Which direction is he facing now ? **(Hotel Management, 1996)**
(a) South (b) South-west (c) West (d) South-east

4. I am facing east. I turn 100° in the clockwise direction and then 145° in the anticlockwise direction. Which direction am I facing now ?

(a) East (b) North-east (c) North (d) South-west

(Hotel Management, 1998)

5. Deepak starts walking straight towards east. After walking 75 metres, he turns to the left and walks 25 metres straight. Again he turns to the left, walks a distance of 40 metres straight, again he turns to the left and walks a distance of 25 metres. How far is he from the starting point ?

(a) 25 metres (b) 50 metres (c) 115 metres
(d) 140 metres (e) None of these

6. Kishenkant walks 10 kilometres towards North. From there, he walks 6 kilometres towards South. Then, he walks 3 kilometres towards East. How far and in which direction is he with reference to his starting point ?

(M.BJV. 1998)

(a) 5 kilometres West (b) 5 kilometres North-east
(c) 7 kilometres East (d) 7 kilometres West

7. A man leaves for his office from his house. He walks towards East. After moving a distance of 20 m, he turns towards South and walks 10 m. Then he walks 35 m towards the West and further 5 m towards the North. He then turns towards East and walks 15 m. What is the straight distance in metres between his initial and final positions ?

(a) 0 (b) 5 (c) 10
(d) Cannot be determined (e) None of these

8. Gaurav walks 20 metres towards North. He then turns left and walks 40 metres. He again turns left and walks 20 metres. Further, he moves 20 metres after turning to the right. How far is he from his original position ?

(a) 20 metres (b) 30 metres (c) 50 metres
(d) 60 metres (e) None of these

(Bank P.O. 1997)

9. Radha moves towards South-east a distance of 7 km, then she moves towards West and travels a distance of 14 m. From here, she moves towards North-west a distance of 7 m and finally she moves a distance of 4 m towards East and stood at that point. How far is the starting point from where she stood ?

(a) 3 m (b) 4 m (c) 10 m (d) 11 m

a. Tax & Central Excise, 1995)

10. Gopal starts from his house towards West. After walking a distance of 30 metres, he turned towards right and walked 20 metres. He then turned left and moving a distance of 10 metres, turned to his left again and walked 40 metres. He now turns to the left and walks 5 metres. Finally he turns to his left. In which direction is he walking now ?

(a) North (b) South (c) East (d) South-west (e) West

11. A rat runs 20' towards East and turns to right, runs 10' and turns to right, runs 9' and again turns to left, runs 5' and then turns to left, runs 12' and finally turns to left and runs 6'. Now, which direction is the rat facing ?

(a) East (b) West (c) North (d) South

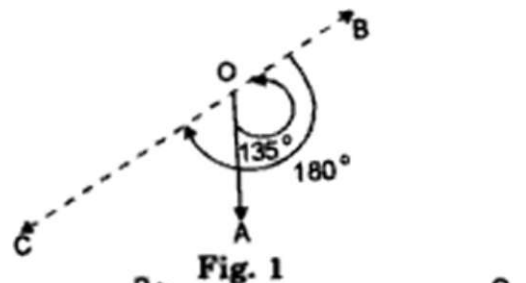
(Assistant Grade, 1996)

12. A girl leaves from her home. She first walks 30 metres in North-west direction and then 30 metres in South-west direction. Next, she walks 30 metres in South-east direction. Finally, she turns towards her house. In which direction is she moving ?
 (a) North-east (b) North-west (c) South-east
 (d) South-west (e) None of these
13. Saiyeev walks 10 metres towards the South. Turning to the left, he walks 20 metres and then moves to his right. After moving a distance of 20 metres, he turns to the right and walks 20 metres. Finally, he turns to the right and moves a distance of 10 metres. How far and in which direction is he from the starting point ?
 (a) 10 metres North (b) 20 metres South (c) 20 metres North
 (d) 10 metres South (e) None of these
14. Kashish goes 30 metres North, then turns right and walks 40 metres, then again turns right and walks 20 metres, then again turns right and walks 40 metres. How many metres is he from his original position ?
 (a) 0 (b) 10 (c) 20 (d) 40 (e) None of these
15. I am facing South. I turn right and walk 20 m. Then I turn right again and walk 10 m. Then I turn left and walk 10 m and then turning right walk 20 m. Then I turn right again and walk 60 m. In which direction am I from the starting point ?
(Hotel Management, 1998)
 (a) North (b) North-west (c) East (d) North-east
16. A man walks 30 metres towards South. Then, turning to his right, he walks 30 metres. Then, turning to his left, he walks 20 metres. Again, he turns to his left and walks 30 metres. How far is he from his initial position ?
 (a) 20 metres (b) 30 metres (c) 60 metres
 (d) 80 metres (e) None of these
17. Rohit walked 25 metres towards South. Then he turned to his left and walked 20 metres. He then turned to his left and walked 25 metres. He again turned to his right and walked 15 metres. At what distance is he from the starting point and in which direction ?
(Bank P.O. 1996)
 (a) 35 metres East (b) 35 metres North (c) 40 metres East
 (d) 60 metres East (e) None of these
18. Starting from a point P, Sachin walked 20 metres towards South. He turned left and walked 30 metres. He then turned left and walked 20 metres. He again turned left and walked 40 metres and reached a point Q. How far and in which direction is the point Q from the point P ?
(Bank P.O. 1992)
 (a) 20 metres West (b) 10 metres East (c) 10 metres West
 (d) 10 metres North (e) None of these
19. Ramakant walks northwards. After a while, he turns to his right and a little further to his left. Finally, after walking a distance of one kilometre, he turns to his left again. In which direction is he moving now ?
 (a) North (b) South (c) East (d) West
20. A man walks 1 km towards East and then he turns to South and walks 5 km. Again he turns to East and walks 2 km, after this he turns to North and walks 9 km. Now, how far is he from his starting point ?
(M.B.A. 1998)
 (a) 3 km (b) 4 km (c) 5 km (d) 7 km

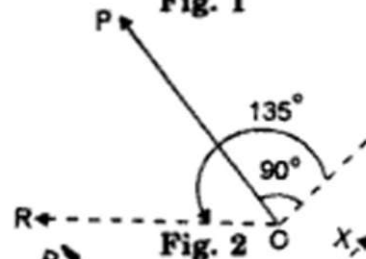
21. travelled from a point X straight to Y at a distance of 80 metres. He turned right and walked 50 metres, then again turned right and walked 70 metres. Finally, he turned right and walked 50 metres. How far is he from the starting point ?
 ia) 10 metres (6) 20 metres (c) 50 metres
 (d) 70 metres ie) None of these
22. Laxman went 15 kms to the west from my house, then turned left and walked 20 kms. He then turned East and walked 25 kms and finally turning left covered 20 kms. How far was he from his house ?
 ia) 5 kms (6) 10 kms (c) 40 kms (d) 80 kms
23. From his house, Lokesh went 15 kms to the North. Then he turned West and covered 10 kms. Then, he turned South and covered 5 kms. Finally, turning to East, he covered 10 kms. In which direction is he from his house ?
 (a) East ib) West (c) North (d) wSouth
 (C M 1996)
24. Going 50 m to the South of her house, Radhika turns left and goes another 20 m. Then, turning to the North, she goes 30 m and then starts walking to her house. In which direction is she walking now ?
 (o) North-west (6) North (c) South-east id) East
25. A walks 10 metres in front and 10 metres to the right. Then every time turning to his left, he walks 5, 15 and 15 metres respectively. How far is he now from his starting point ?
 (a) 5 metres (6) 10 metres (c) 15 metres
 id) 20 metres (e) 23 metres
26. Rasik walks 20 m North. Then he turns right and walks 30 m. Then he turns right and walks 35 m. Then he turns left and walks 15 m. Then he again turns left and walks 15 m. In which direction and how many metres away is he from his original position ?
 (a) 15 metres West (6) 30 metres East
 (c) 30 metres West id) 45 metres East
 (C.B.I. 1997)
27. A child is looking for his father. He went 90 metres in the East before turning to his right. He went 20 metres before turning to his right again to look for his father at his uncle's* place 30 metres from this point. His father was not there. From here he went 100 metres to the North before meeting his father in a street How far did the son meet his father from the starting point ?
 ia) 80 metres ib) 100 metres ic) 140 metres id) 260 metres
 (L Tax & Central Excise, 1996)
28. The door of Aditya's house faces the East. From the back side of his house, he walks straight 50 metres, then turns to the right and walks 50 metres again. Finally, he turns towards left and stops after walking 25 metres. Now, Aditya is in which direction from the starting point ?
 ia) South-east ib) North-east (c) South-west id) North-west

ANSWERS

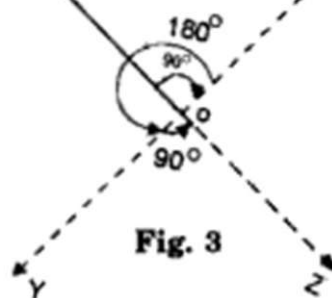
1. (d): As shown in Fig. 1, the man initially faces in the direction OA. On moving 135° anticlockwise, he faces in the direction OB. On further moving 180° clockwise, he faces in the direction OC, which is South-west.



2. (6): As shown in Fig. 2, the man initially faces in the direction OP. On moving 90° clockwise, the man faces in the direction OQ. On further moving 135° anticlockwise, he faces in the direction OR, which is West.



3. (<d): As shown in Fig. 3, the man initially faces in the direction OP. On moving 90° clockwise, he faces in the direction OX. On further moving 180° anticlockwise, he faces in the direction OY. Finally, on moving 90° anticlockwise, he faces in the direction OZ, which is South-east.



4. (6): As shown in Fig. 3, the man initially faces towards east i.e., in the direction OA. On moving 100° clockwise, he faces in the direction OB. On further moving 145° clockwise, he faces in the direction OC. Clearly, OC makes an angle of $(145^\circ - 100^\circ)$ i.e. 45° with OA and as such points in the direction North-east.

145
8 * Fig. 4

5. (e): The movements of Deepak are as shown in Fig. 5.

Clearly, $EB \ll DC = 40$ m.

$\therefore$ Deepak's distance from the starting point A
 $= (AB - EB) * (75 - 40)$ m = 35 m.

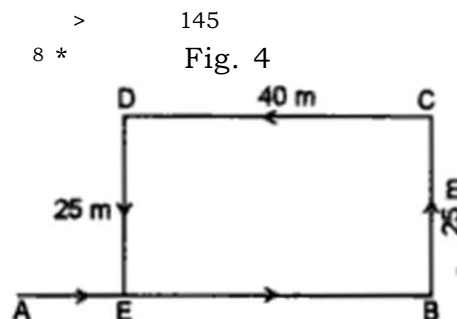


Fig. 5

6. (6): The movements of Kishenkant are as shown in Fig. 6 (A to B, B to C and C to D).

$AC = (AB - BC)$ a $(10 - 6)$ km = 4 km.

Clearly, D is to the North-east of A.

Kishenkant's distance from starting point A
 $= AD = \sqrt{AC^2 + CD^2} = \sqrt{4^2 + 3^2} = \sqrt{25} = 5$ km.

So, Kishenkant is 5 km to the North-east of his starting point.

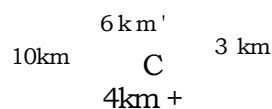


Fig. 6

7. (6): The movements of the man from A to F are as shown in Fig. 7.

Clearly, $DC = AB + EF$.

F is in line with A.

Also, $AF = (BC - DE) = 5 \text{ m}$.

So, the man is 6 metres away from his initial position.

8. (d): The movements of Gaurav are as shown in Fig- 8.

Clearly, Gaurav's distance from his initial position $P = PX = (PS + SX) = (QR + SX)$

$$= (40 + 20) \text{ m} = 60 \text{ m}.$$

9. (c): The movements of Radha are as shown in Fig. 9.

Clearly, Radha's distance from the starting point $O = OD = (OC - CD)$

$$= (AB - CD) = (14 - 4) \text{ m} = 10 \text{ m}.$$

10. (a): The movements of Gopal are as shown in Fig. 10 from A to G.

Clearly, Gopal is finally walking in the direction FG i.e. North. -

11. (c): The movements of the rat from A to G are as shown in Fig. 11.

Clearly, it is finally walking in the direction FG i.e. North.

12. (a): The movements of the girl are as shown in Fig. 12 (A to B, B to C, C to D, D to A).

Clearly, she is finally moving in the direction DA i.e. North-east.

13. (6): The movements of Saryeev from A to F are as shown in Fig. 13.

Clearly, Sanjeev's distance from starting point A

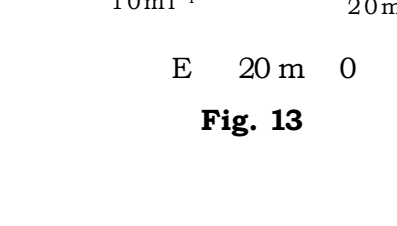
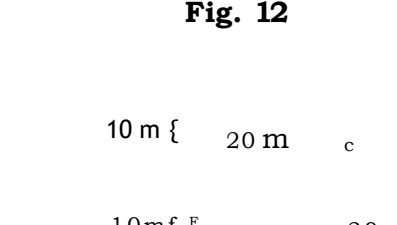
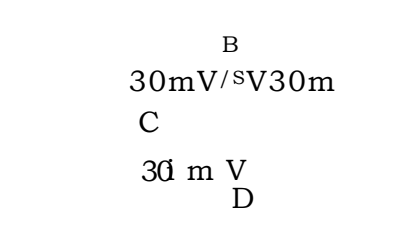
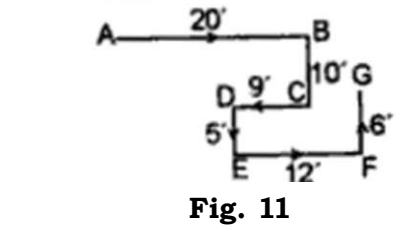
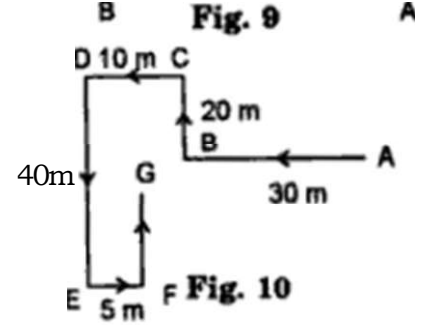
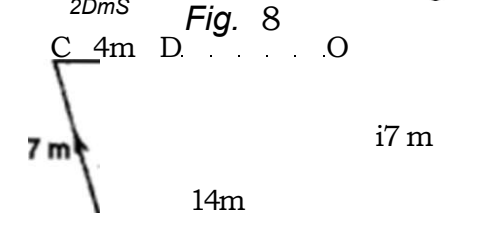
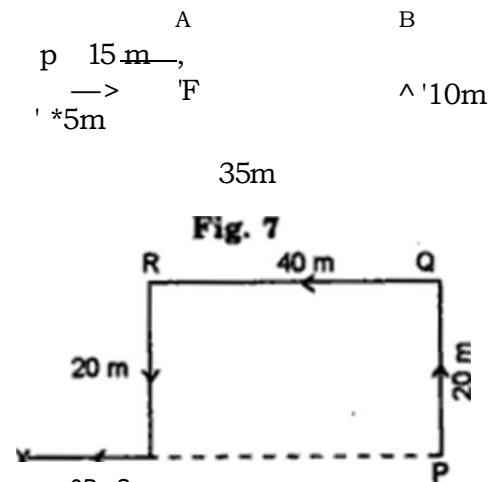
$$= AF = (AB - BF)$$

$$= AB + (BE - EF) = AB + (CD - EF)$$

$$= [10 + (20 - 10)] - (10 + 10) \text{ m} = 20 \text{ m}.$$

Also, F lies to the South of A.

So, Sanjeev is 20 metres to the south of his starting point.



14. (b): The movement* of Kashish are as shown in Fig. 14 (A to B, B to C, C to D, D to E)

Hashish's distance from his original position A = AE = (AB - BE) = (AB - CD) = (30 - 20) m = 10 m.

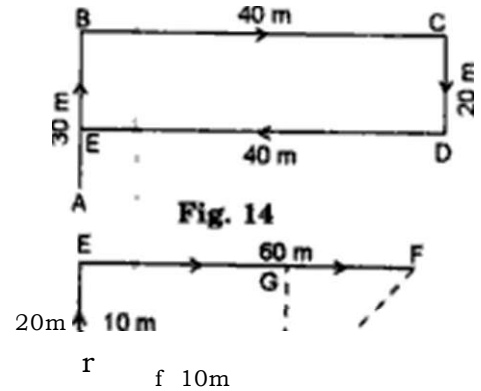


Fig. 14

15. (d): The movements of the person are from A to F, as shown in Fig. 15. Clearly, the final position is F which is to the North-east of the starting point A.

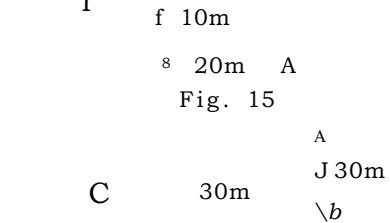


Fig. 15

16. (*): The movements of the man are as shown in Fig. 16.

Man's distance from initial position A = AE - (AB + BE) = (AB - CD) = (30 - 20) m = 10 m.

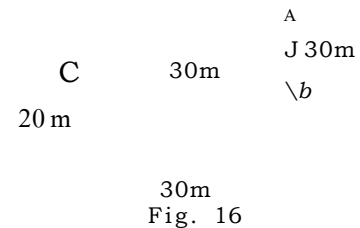


Fig. 16

17. (a): The movements of Rohit are as shown in Fig. 17.

∴ Rohit's distance from starting point A = AE - (AD + DE) = (BC + DE) = (20 + 15) m = 35 m. Also, E is to the East of A

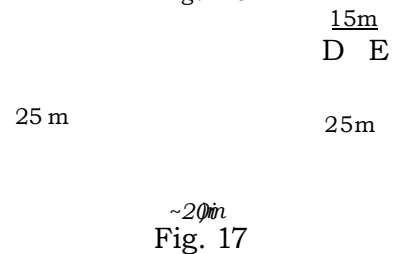


Fig. 17

18. (c): The movements of Sachin are as shown in Fig. 18 (P to B, B to C, C to D and D to Q). Clearly, distance of Q from P

= PQ - (DQ - PD) = (DQ - BC) = (40 - 30) m = 10 m.

Also, Q is to the West of P. Q is 10 m West of P.

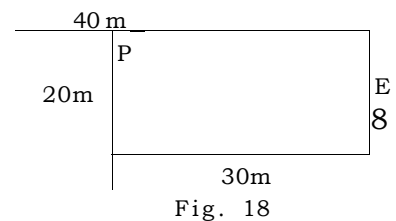


Fig. 18

19. (d): The movements of Ramakant are as shown in Fig. 19,

Clearly, he is finally walking in the direction DE i.e., West

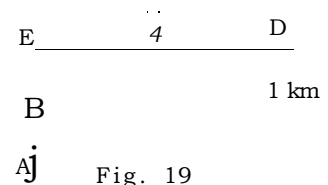


Fig. 19

20. (c): The movements of the man are as shown in Fig. 20 (A to B, B to C, C to D, D to E).

Clearly, DF = BC - 5 km

EF = (DE - DF) = (9 - 5) km = 4 km.

BF = CD = 2 km.

AF = AB + BF = AB + CD = (1 + 2) km = 3 km.

Man's distance from starting point A

= AE = $\sqrt{AF^2 + EF^2} = \sqrt{3^2 + 4^2}$

= $\sqrt{25} = 5$ km.

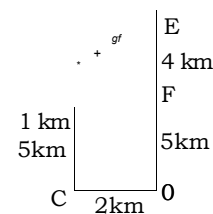


Fig. 20

21. (a): The movements of Rty are as shown in Fig. 21 (X to Y, Y to A, A to B, B to C).

$$\begin{aligned}\text{Raj's distance from the starting point X} &= XC = (XY - YC) \\ &= (XY - BA) = (80 - 70) \text{ m} = 10 \text{ m}.\end{aligned}$$

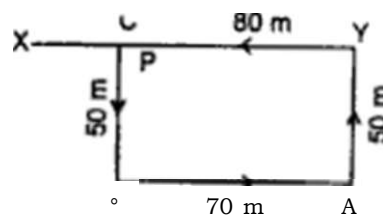


Fig. 21

22. (6): The movements of Laxman are as shown in Fig. 22.

$$\begin{aligned}\therefore \text{Laxman's distance from his house at A} &- AE = (BE - BA) \\ &= (CD - BA) = (25 - 15) \text{ m} = 10 \text{ m}.\end{aligned}$$

B 15 m a

20 m 20 m



Fig. 22

23. (c): The movements of Lokesh are as shown in Fig. 23 (A to B, B to C, C to D and D to E). Clearly, his final position is E which is to the North of his house at A.

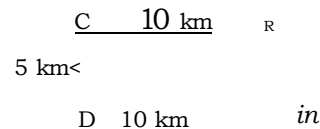


Fig. 23

24. (o): The movements of Radhika are as shown in Fig. 24 (A to B, B to C, C to D and D to A). Clearly, she is finally moving in the direction DA i.e. North-west.

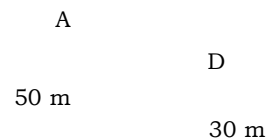


Fig. 24

25. (a): The movements of A are as shown in Fig. 25 (O to P, P to Q, Q to R, R to S and S to T). Since TS = OP + QR, so T lies in line with O.

$$\begin{aligned}\text{A's distance from the starting point O} &\bullet OT = (RS - PQ) = (15 - 10) \text{ m} = 5 \text{ m}.\end{aligned}$$

Fig. 25

26. (d): The movements of Rasik from A to F are as shown in Fig. 26.

Since CD = AB + EF, so F lies in line with A.

$$\begin{aligned}\therefore \text{Rasik's distance from original position A} &= AF = (AG + GF) \\ &= (BC + DE) = (30 + 15) \text{ m} = 45 \text{ m}.\end{aligned}$$

Also, F lies to the east of A.

20

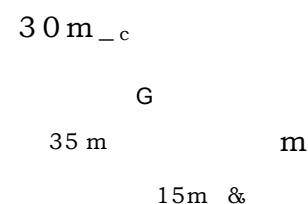


Fig. 26

27. (6): The movements of the child from A to E are as shown in Fig. 27.

Clearly, the child meets his father at E.

$$\begin{aligned}\text{Now, } AF &= (AB - FB) \\ &= (AB - DC) = (90 - 30) \text{ m} = 60 \text{ m}.\end{aligned}$$

$$\begin{aligned}EF &= (DE - DF) = (DE - BC) \\ &= (100 - 20) \text{ m} = 80 \text{ m}.\end{aligned}$$

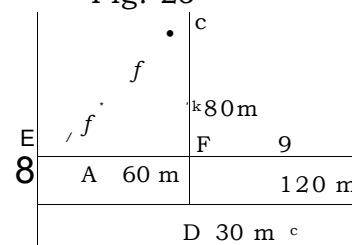
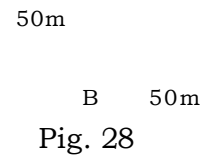


Fig- 27

$$\text{Required distance} = \sqrt{AE^2 + EF^2} = \sqrt{(60)^2 + (80)^2} = 100 \text{ m.}$$

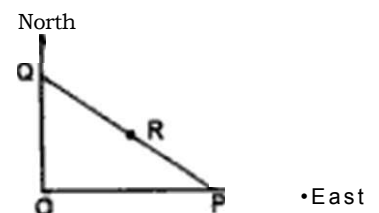
28. *id*): Since Aditya's house face* toward* East and he walks from backside of his house, it means that he starts walking towards West. Thus, the movement* of Aditya are as shown in Fig. 28 (A to B, B to C, C to D). Clearly, Aditya's final position is D which is to the North-west of the starting point A.



EXERCISE 88

- Two buses start from the opposite points of a main road, 150 kms apart. The first bus runs for 25 kms and takes a right turn and then runs for 15 kms. It then turns left and runs for another 25 kms and takes the direction back to reach the main road. In the meantime, due to a minor breakdown, the other bus has run only 35 kms along the main road. What would be the distance between the two buses at this point ? (L Tax A Central Excise, 1996)
 (a) 65 kms (b) 75 kms (c) 80 kms (d) 85 kms
- X and Y start moving towards each other from two places 200 m apart. After walking 60 m, B turns left and goes 20 m, then he turns right and goes 40 m. He then turns right again and comes back to the road on which he had started walking. If A and B walk with the same speed, what is the distance between them now ?
 (a) 20 m (b) 30 m (c) 40 m (d) 50 m
- If A is to the south of B and C is to the east of B, in what direction is A with respect to C ?
 (a) North-east (b) North-west (c) South-east
 (d) South-west (e) None of these
- A is 40 m South-west of B. C is 40 m South-east of B. Then, C is in which direction of A ? (Assistant Grade, 1997)
 (a) East (b) West (c) North-east (d) South
- There are four towns P, Q, R and T. Q is to the South-west of P, R is to the east of Q and south-east of P, and T is to the north of R in line with QP. In which direction of P is T located ?
 (a) South-east (b) North (c) North-east (d) East

6. In the given figure. P is 300 km eastward of O and Q is 400 km north of O. R is exactly in the middle of Q and P. The distance between Q and R is (I.A.S. 1997)



- (a) 250 km (b) 250 km (c) 300 km (d) 350 km

7. Ravi wants to go to the university. He starts from his home which is in the East and comes to a crossing. The road to the left ends in a theatre, straight ahead is the hospital. In which direction is the university ? (Railways, 1998)
(a) North (b) South (c) East **(d)** West
8. Of the six members of a panel sitting in a row, A is to the left of D, but on the right of E. C is on the right of X, but is on the left of B who is to the left of F. Which two members are sitting right in the middle ?
 (a) A and C (b) C and B (c) D and B (d) D and C
9. A, B, C and D are playing cards. A and B are partners. D faces towards North. If A faces towards West, then who faces towards South ?
 (a) B (b) C (c) D (d) Data inadequate
10. P, Q, R and S are playing a game of carrom. P, R and S, Q are partners. S is to the right of R who is facing west. Then, Q is facing (Hotel Management, 1992)
 (a) North (b) South (c) East (d) West
11. The town of Paranda is located on Green Lake. The town of Akram is west of Paranda. Tokhada is east of Akram but west of Paranda. Kakran is east of Bopri but west of Tokhada and Akram. If they are all in the same district, which town is the farthest west ?
 (a) Paranda (b) Kakran (c) Akram (d) Bopri
12. Five boys are standing in a row facing East. Deepak is to the left of Sameer, Tushar and Shailendra. Sameer, Tushar and Shailendra are to the left of Sushil. Shailendra is between Sameer and Tushar. If Tushar is fourth from the left, how far is Sameer from the right ?
 (a) First (b) Second (c) Third (d) Fourth (e) Fifth
- Directions (Questions 13-14): **Study the information given below carefully and answer the questions that follow :** (Bank P.O. 1997)
- On a playing ground, Dinesh, Kunal, Nitin, Atul and Prashant are standing as described below facing the North.
- (i)** Kunal is 40 metres to the right of Atul.
 - (ii)** Dinesh is 60 metres to the south of Kunal.
 - (iii)** Nitin is 25 metres to the west of Atul.
 - (iv)** Prashant is 90 metres to the north of Dinesh.
13. Who is to the north-east of the person who is to the left of Kunal ?
(a) Dinesh **(b)** Nitin (c) Atul
(d) Either Nitin or Dinesh **(e)** None of these
14. If a boy walks from-Nitin, meets Atul followed by Kunal, Dinesh and then Prashant, how many metres has he walked if he has travelled the straight distance all through ?
(a) 155 metres (b) 185 metres **(c)** 215 metres
(d) 245 metres (e) None of these
15. Two ladies and two men are playing cards and are seated at North, East, South and West of a table. No lady is facing East. Persons sitting opposite to each other are not of the same sex. One man is facing South. Which directions are the ladies facing ?
 (a) East and West (b) South and East (c) North and East
 (d) North and West (e) None of these

16. The post office is to the east of the school while my house is to the south of the school. The market is to the north of the post office. If the distance of the market from the post office is equal to the distance of my house from the school, in which direction is the market with respect to my school ?
 (a) North (b) East (c) North-east (d) South-west
17. Lokesh's school bus is facing North when it reaches his school. After starting from Lokesh's house, it turns right twice and then left before reaching the school. What direction was the bus facing when it left the bus stop in front of Lokesh's house ?
 (a) North (b) South (c) East (d) West (e) None of these
18. I start from my home and go 2 km straight. Then, I turn towards my right and go 1 km. I turn again towards my right and go 1 km again. If I am north-west from my house, then in which direction did I go in the beginning ?
 (a) North (b) South (c) East (d) West (e) South-east
19. After walking 6 km, I turned right and covered a distance of 2 km, then turned left and covered a distance of 10 km. In the end, I was moving towards the north. From which direction did I start my journey ?
 (a) North (b) South (c) East (d) West

(I. Tax & Central Excise, 1994)

20. A postman was returning to the post office which was in front of him to the north. When the post office was 100 metres away from him, he turned to the left and moved 50 metres to deliver the last letter at Shantivilla. He then moved in the same direction for 40 metres, turned to his right and moved 100 metres. How many metres was he away from the post office ?
 (a) 0 (b) 90 (c) 150 (d) 100 (e) None of these
21. A boy rode his bicycle northwards, then turned left and rode one km and again turned left and rode 2 km. He found himself exactly one km west of his starting point. How far did he ride northwards initially ? **(Assistant Grade, 1994)**
 (a) 1 km (b) 2 km (c) 3 km (d) 5 km
22. If 'South-east' is called 'East', 'North-west' is called 'West' 'South-west' is called 'South' and so on, what will 'North' be called ?
 (a) East (b) North-east (c) North-west (d) South (e) None of these
23. If South-east becomes North, North-east becomes West and so on, what will West become ? **(Assistant Grade, 1998)**
 (a) North-east (b) North-west (c) South-east
 (d) South-west (e) South
24. A direction pole was situated on the crossing. Due to an accident the pole turned in such a manner that the pointer which was showing East, started showing South. One traveller went to the wrong direction thinking it to be West. In what direction actually he was travelling ? **(M.B.A 1998)**
 (a) North (b) South (c) East (d) West
25. A watch reads 4.30. If the minute hand points East, in what direction will the hour hand point ?
 (a) North (b) North-west (c) South-east
 (d) North-east (e) None of these

26. A clock is so placed that at 12 noon its minute hand points towards north-east. In which direction does its hour hand point at 1.30 p.m. ?

(a) North (b) South (c) East (d) West

(Hotel Management, 1995)

27. If the above clock is turned through an angle of 135° in an anticlockwise direction, in which direction will its minute hand point at 8.45 p.m. ?

(a) North (b) South (c) East (d) West

Directions (Questions 28 to 32) : Read the following information carefully and answer the questions given below it : (Bank P.O. 1995)

- ii) Six flats on a floor in two rows facing North and South are allotted to P, Q, R, S, T and U.
(ii) Q gets a North facing flat and is not next to S.
(iii) S and U get diagonally opposite flats.
(iv) R, next to U, gets a South facing flat and T gets a North facing flat.
28. Which of the following combinations get South facing flats ?
(a) QTS (b) UPT (c) URP (d) Data inadequate (e) None of these
29. Whose flat is between Q and S ?
(a) T (b) U (c) R (d) P (e) Data inadequate
30. If the flats of T and P are interchanged, whose flat will be next to that of U ?
(a) P (b) Q (c) R (d) T (e) None of these
31. The flats of which of the other pairs than SU, is diagonally opposite to each other ?
ia) QP (b) QR (c) PT (d) TS (e) None of these
32. To arrive at the answers to the above questions, which of the following statements can be dispensed with ?
(a) None (b) (i) only (c) (ii) only
id) (iii) only ie) None of these
33. One morning after sunrise, Gopal was standing facing a pole. The shadow of the pole fell exactly to his right. Which direction was he facing ?
(a) South (b) East (c) West
(d) Data inadequate (e) None of these **(Bank P.O. 1997)**
34. One morning after sunrise, Reeta and Kavita were talking to each other face to face at Tilak Square. If Kavita's shadow was exactly to the right of Reeta, which direction Kavita was facing ? **(Bank P.O. 1998)**
(a) North (b) South (c) East
(d) Data inadequate (e) None of these
35. One morning after sunrise, Vikram and Shailesh were standing in a lawn with their backs towards each other. Vikram's shadow fell exactly towards left hand side. Which direction was Shailesh facing ?
(a) East (b) West (c) North (d) South
36. One evening before sunset two friends Sumit and Mohit were talking to each other face to face. If Mohit's shadow was exactly to his right side, which direction was Sumit facing ? **(Bank P.O. 1997)**
ia) North (b) South (c) West
id) Data inadequate ie) None of these

37. Anuj started walking positioning his back towards the sun. After sometime, he turned left, then turned right and then towards the left again. In which direction is he going now ? (I. Tax & Central Excise, 1994)
- (a) North or South (b) East or West
(c) North or West (d) South or West

Directions (Questions 38 to 42) : The following questions are based on the diagram given below showing four persons stationed at the four corners of a square piece of plot as shown. <

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38. A starts crossing the field diagonally. After walking half the distance, he turns right, walks some distance and turns left. Which direction is A facing now ?
(a) North-east (b) North-west (c) North (d) South-east (e) South-west
39. From the original position given in the above figure, A and B move one arm length clockwise and then cross over to the corner diagonally opposite; C and D move one arm length anti-clockwise and cross over the corner diagonally opposite. The original configuration ADBC has now changed to
(a) CBDA (b) BDAC (c) DACB (d) ACBD (e) BCAD
40. From the original position, B and D move one and a half length of sides clockwise and anticlockwise respectively. Which one of the following statements is true ?
(a) B and D are both at the midpoint between A and C.
(b) D is at the midpoint between A and C, and B at the corner originally occupied by C.
(c) B is at the midpoint between A and C, and D at the corner originally occupied by A.
(d) B and D are both at the midpoint between A and D.
(e) B is at the midpoint between A and C, and D at the midpoint between original position of B and C.
41. From the positions in original figure, C and A move diagonally to opposite corners and then one side each clockwise and anticlockwise respectively. B and D move two sides each clockwise and anticlockwise respectively. Where is A now ?
(a) At the north-west corner (b) At the north-east corner
(c) At the south-east corner (d) At the south-west corner
(e) Midway between original position of B and D
42. After the movements given in Q. 41 above, who is at the north-west corner ?
(a) A (b) B (c) C (d) D (e) None of these
43. A square field ABCD of side 90 m is so located that its diagonal AC is from north to south and the corner B is to the west of D. Rohan and Rahul start walking along the sides from B and C respectively in the clockwise and anti-clockwise directions with speeds of 8 km/hr and 10 km/hr. Where shall they cross each other the second time ? (Hotel Management, 1998)

- (a) On AD at a distance of 30 m from A
- (b) On BC at a distance of 10 m from B
- (c) On AD at a distance of 30 m from D
- (d) On BC at a distance of 10 m from C

ANSWERS

1. (a): Let X and Y be two buses

Bus X travels along the path

PA, AB, BC, CD

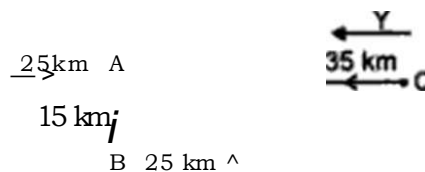
Now, $AD \gg BC \gg 25$ km.

So, $PD = PA + AD = 50$ km.

Bus Y travels 35 km upto E.

Distance between two buses $\Rightarrow PQ - (PD + QE)$

$$= [150 - (50 + 35)] = 65 \text{ km.}$$



2. (c): Clearly Y moves 60 m from Q

upto A, then 20 m upto B, 40 m

upto C and then upto D.

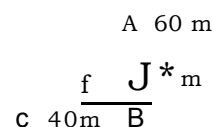
So, $AD = BC = 40$ m.

$QD = (60 + 40) \text{ m} = 100$ m.

Since A and B travel with the same speed, A will travel the same speed along the horizontal as B travels in the same time *i.e.* $(60 + 20 + 40 + 20) = 140$ m.

So, X travels 140 m upto A

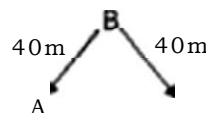
Distance between X and Y = $AD = (100 - 60) \text{ m} = 40$ m.



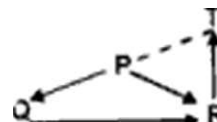
3. (d) : Clearly comparing the direction of A w.r.t. C in the second diagram with that in the first diagram. A will be south-west of C.



4. (a) : As is clear from the adjoining diagram, C lies to the east of A.



5. (c) : Clearly, the arrangement according to the given directions is as shown. So, T lies to the north-east of P.

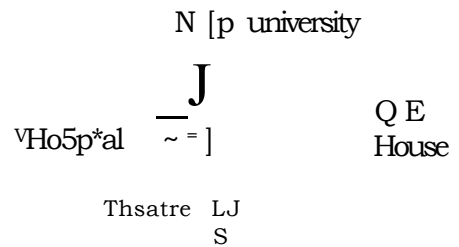


(a): Clearly. $PQ = \sqrt{NOP^2 + OQ^2} = \sqrt{(300)^2 + (400)^2}$
 $= \sqrt{90000 + 160000} \gg 500$ km.

Since R is the midpoint of PQ, so $QR = \frac{1}{2} \times PQ$

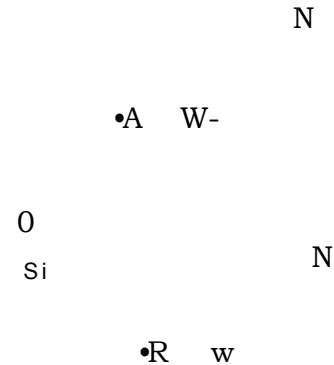
$$= 250 \text{ km.}$$

7. (a) Starting from his house in the East, Ravi moves westwards. Then, the theatre, which is to the left, will be in the South. The hospital, which is straight ahead, will be to the West. So, the Einiveraity will be to the North.



8. (d) A is to the left of D means AD. A is to the right of E means EA. So, the bequence is EAD. C is to the right of X means XC. C is to the left of B means CB. B is to the left of F means BF. So, the sequence is XCBF. Thus, there are two possible arrangements — EADXCBF and XCBFEAD. In the first arrangement, the two members in the middle arc D and C. In the second arrangement, the two members in the middle are B and E. So, from amongst the choices, D and C is the answer.

9. (6): As per the data, D faces North. A faces towards West. So, its partner B will face towards A and hence towards East. So, C who will face D will face towards South.

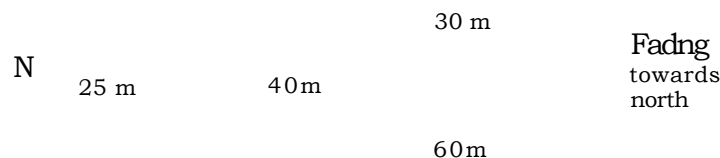


10. (a): As per the given data, R faces towards Went. S is to the right of R. So, S is facing towards South. Thus, Q who is the partner of S, will face towards North.

11. (d): The town of Akram (A) is west of Paranda (P) means A, P. Tokhada (T) is east of Akram but west of Paranda means A, T, P. Kakran (K) is east of Bopri (B) but west of Tokhada and Akram means B, K, A, T. Combining all the arrangements, we get the sequence as B, K, A, T, P. So, farthest west is Bopri.
12. (d) : Deepak (D) is to the left of Sameer (S), Tushar (T) and Shailendra (Sh) means D, S, T, Sh. Sameer, Tushar and Shailendra are to the left of Sushil (Su) means S, T, Sh, Su. Shailendra is between Sameer and Tushar means S, Sh, T. Tushar is fourth from the left means • • • T. Combining all the arrangements, we have D. S, Sh, T, Su. So, Sameer is fourth from the right.

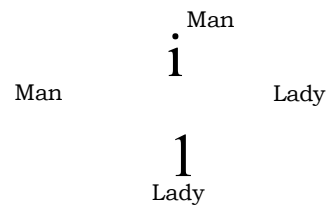
Questions 13-14

Clearly, the arrangement of boys is as shown below :

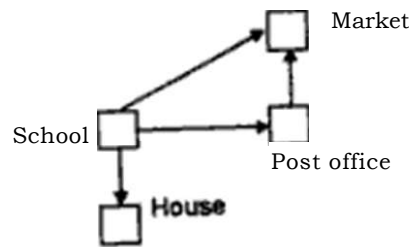


13. (e) Clearly, Atul is to the left of Kunal and Prashant is to the north-east of Atul.
14. (c) Required distance = NA + AK + KD • DP
= (25 + 40 + 60 + 90) - 215 m.

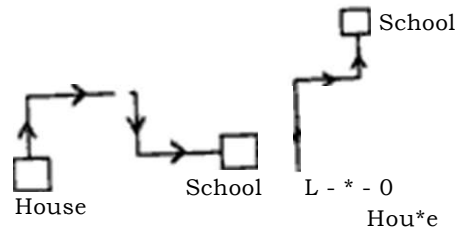
15. W) : No lady is facing east means a man faces east. Persons opposite are not of same sex. So, a woman will be facing west. Again, a man faces south. So, opposite to him will be a woman facing north.



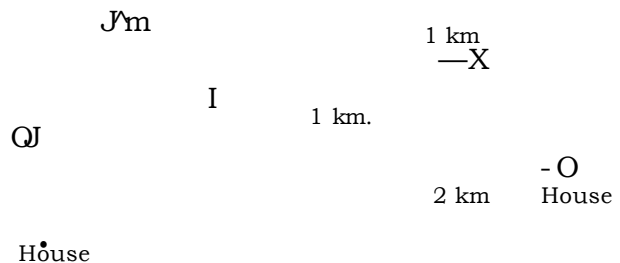
16. (c): The positions of various places are as shown in the diagram. Clearly, the market is to the north-east of school.



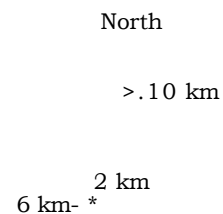
17. (d): In Fig. 1, the route of the bus from Lokesh's house to the school. It is given that the bus faces North on reaching the school. Now, turning Fig. 1 90° anticlockwise, we obtain Fig. 2 which satisfies the specified conditions. It is evident from Fig. 2 that the bus faces west in front of Lokesh's house.



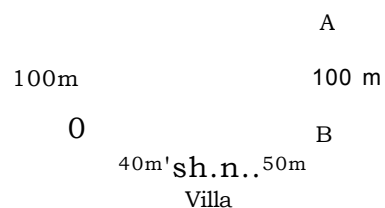
18. (d): Clearly, the route is as shown in Fig. A. It is given that the person is finally to the north-west of his house. Rotating Fig. A 90° anticlockwise, we obtain Fig. B satisfying the specified conditions. It is evident from Fig. B that the direction of walking in the beginning was west.



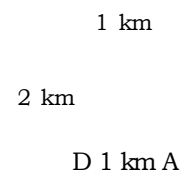
19. (6): Clearly, the route is as shown in the adjoining diagram. Thus, the man started his journey from the South and moved northwards.



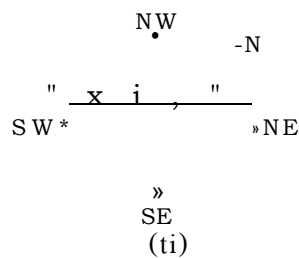
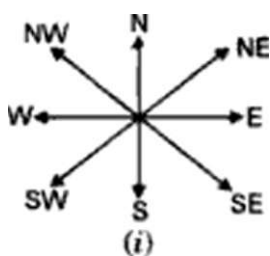
20. (6): Clearly, the route of the postman is as shown. So, at the final point the distance of postman from post office - $EA = BD$
 $z BC \neq CD$
 $\Rightarrow (50 + 40) - 90 \text{ m.}$



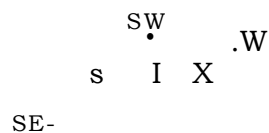
21. (6): Clearly, the boy rode from A to B, then to C and finally upto D. Since D lies to the west of A, so required distance
 $= AB = CD = 2 \text{ km.}$



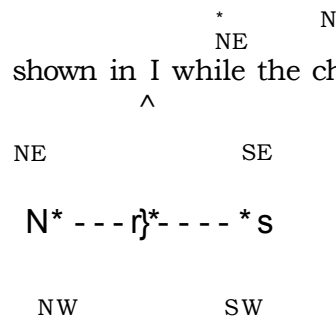
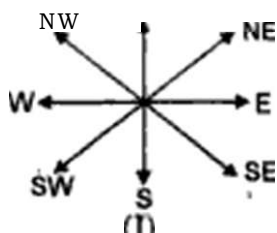
22. (c): In diagram (i) the directions are shown as they actually are. Diagram (ii) is as per the given data. So, comparing the direction of North in (i) with that in (ii), North will be called North-west.



23. (c): Comparing (i) of Q. 22 with the adjoining diagram. West will be called South-east.

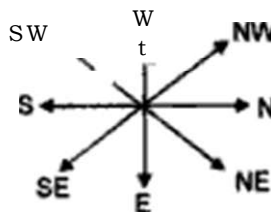


24. (6): The actual positions of the directions are as shown in I while the changed positions of the pointer can be described by II.

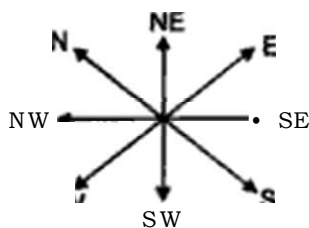


Clearly, the direction of West shown by the pointer in wrong position (Fig. II) is actually South.

25. (rf): Clearly, to show 4.30, the position of the minute and hour hands of the clock will be as shown. So, again as shown, if the minute hand points East, the hour hand will point in the North-east.



26. (c): Clearly, the positions of the minute and hour hands at 12 noon and 1.30 p.m. are as shown in the diagram. So, as shown, the hour hand at 1.30 p.m. points towards the East.



27. (6): The position of the clock on rotating the above clock 135° anticlockwise is as shown in the adjoining diagram. Clearly, the minute hand points towards the South.



Questions 28 to 32

Q gets a North-facing flat and is not next to S means

• • •

S and U get diagonally opposite flats means

B C D
0 • a

R, next to U. gets a South facing flat and T gets a north facing flat means

U R 0
Q T H

So, the arrangement is :__

South facing flats [V] j_R | j > J

North facing flats Q [V] [j T]

^

28. (c): The South facing flats are U; R, P.

29. (a): T's flat is between Q and S.

30. (c): The flat next to **IT** flat is that of R, which remains unchanged if the flats of T and P are interchanged.

31. (a): The diagonally opposite pairs are SU and QP.

32. (a): Clearly, all the statements are necessary to answer the given questions.

33. (a): Sun rises in the east in the morning. So, in morning, the shadow falls towards the west. Now, Copal's shadow falls to the right. So, W « he is standing, facing South.

34. (a): In morning, sun rises in the east and so any shadow falls towards the west. Now, Kavita's shadow falls to the right of Reeta. I
Hence, Reeta is facing South and Kavita is facing North. **Kavita's Shadow** > »

36. (d): Since Vikram's shadow fell towards left, therefore, Vikram is facing North. So, Shailesh standing with his back towards Vikram, will be facing South.

36. (6): In the evening, sun is in the west and so the shadows fall towards east. Now, since Mohit's shadow fell towards right, therefore, Mohit is facing North. So, Sumit standing face to face with Mohit, was facing South.

37. (a): Clearly, there are two possible movement* of Anuj as shown below :

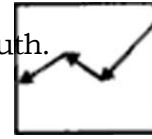
North



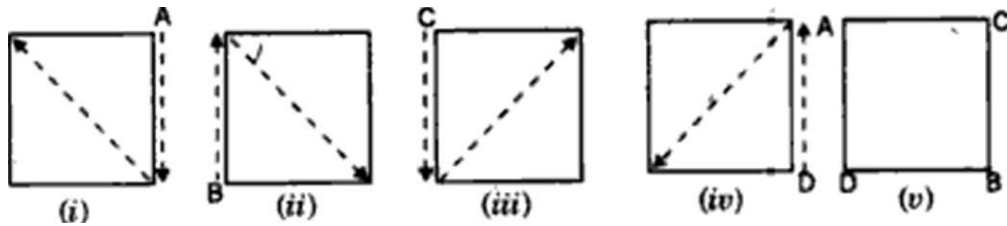
D > J

South

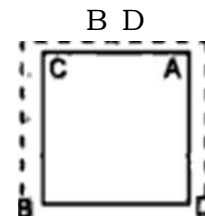
38. (b): Clearly, the route of A is as shown. Thus, Amy is finally moving towards either North or South. Comparing it with the given diagram, the direction of A will be north-west



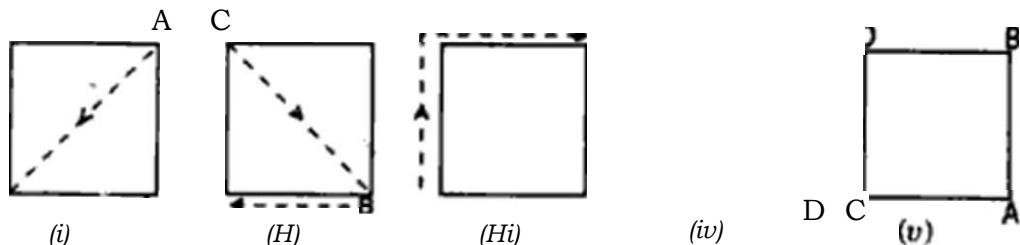
39. (a): Clearly, (i), (ii), (iii) and (iv) show the movements of A, B, C and D respectively while the new arrangement so obtained is shown in (v). So, the configuration changes to CBDA.



40. (o): The movements of B and D are clearly shown in the adjoining diagram. So, statement (a) is true.



41. (d): The movements of A, C, B and D are shown in figures (i), (ii), (iii) and (iv) respectively. The final configuration is shown in (v). Comparing (v) with the given diagram, A is in the south-west corner.



42. (c): Clearly, C is at the north-west corner

43. (d) : Clearly, the arrangement is as shown in the adjoining diagram.

Rohan's speed = 8 km/hr

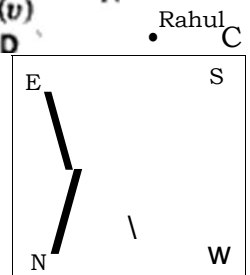
$$\frac{8000}{60 \times 60} \text{ m/sec} = \frac{20}{9} \text{ m/sec}$$

Rahul's speed = 10 km/hr

$$\frac{10000}{60 \times 60} \text{ m/sec} = \frac{25}{9} \text{ m/sec}$$

Since Rohan and Rahul are moving in opposite directions, so they together cover a distance of $\left| \frac{20}{9} + \frac{25}{9} \right|$ i.e., $\frac{45}{9}$ or 5 metres in one second.

To meet at a point, they together have to cover distance (CD + DA + AB) i.e. 270 m.



Rohan

Now, 5 metres is covered in 1 second

So, 270 m will be covered in $\frac{270}{5} = 54$ seconds.

y Now, distance covered by Rohan in 54 seconds = $5 \times 54 = 270$ m.

Distance covered by Rahul in 54 seconds = $3 \times 54 = 162$ m.

Thus, Rohan and Rahul meet for the first time on AD at a point 30 m from A and 60 m from D. Now, to meet again, Rohan and Rahul will have to complete one full round i.e. together move a distance of 360 m.

5 metres is covered by both together in 1 second.

Thus, 360 m will be covered by both in $\frac{360}{5} = 72$ seconds.

Now, distance covered by Rohan in 72 seconds = $5 \times 72 = 360$ m.

Distance covered by Rahul in 72 seconds = $3 \times 72 = 216$ m.

Thus, Rohan and Rahul meet on BC at a point 10 m from C and 80 m from B.